

Part1:Generate Client Certificates

- Verify that all certificate files exist in the `certs/` directory:

File	Purpose	Created?
certs/ca.pem	Certificate Authority	[✓] Yes
certs/ca-key.pem	CA private key	[✓] Yes
certs/server.pem	Broker certificate	[✓] Yes
certs/server-key.pem	Broker private key	[✓] Yes
certs/device-001.pem	Device 001 certificate	[✓] Yes
certs/device-001-key.pem	Device 001 private key	[✓] Yes
certs/device-002.pem	Device 002 certificate	[✓] Yes
certs/device-002-key.pem	Device 002 private key	[✓] Yes
certs/device-003.pem	Device 003 certificate	[✓] Yes
certs/device-003-key.pem	Device 003 private key	[✓] Yes

- Verify that all certificate files exist in the `certs/` directory:

File	Purpose	Created?
<code>certs/ca.pem</code>	Certificate Authority	
<code>certs/ca-key.pem</code>	CA private key	
<code>certs/server.pem</code>	Broker certificate	
<code>certs/server-key.pem</code>	Broker private key	
<code>certs/device-001.pem</code>	Device 001 certificate	
<code>certs/device-001-key.pem</code>	Device 001 private key	
<code>certs/device-002.pem</code>	Device 002 certificate	
<code>certs/device-002-key.pem</code>	Device 002 private key	
<code>certs/device-003.pem</code>	Device 003 certificate	
<code>certs/device-003-key.pem</code>	Device 003 private key	

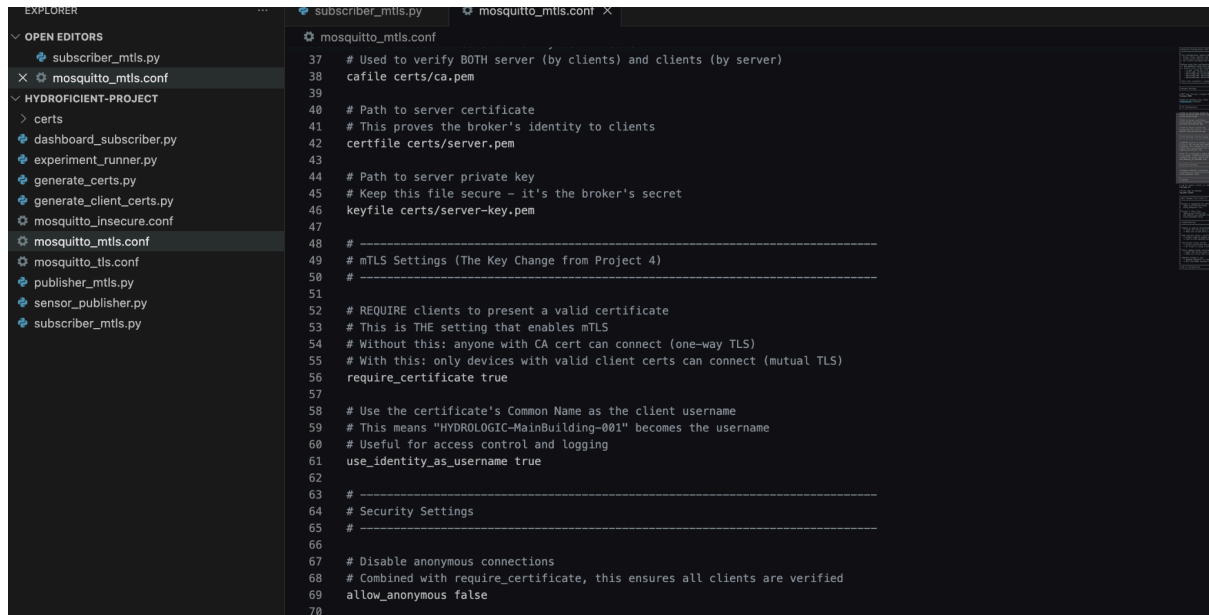
Hydroficient-project — zsh — 87x33

```
Last login: Tue Sep 22 10:21:59 on ttys002
christinekh@christinekhsMBP ~ % cd "/Users/christinekh/Documents/IoT Hydroficient-Extern
n/Week5/Hydroficient-project"
christinekh@christinekhsMBP Hydroficient-project % pwd
/Users/christinekh/Documents/IoT Hydroficient-Extern/Week5/Hydroficient-project
christinekh@christinekhsMBP Hydroficient-project % ls
certs                               mosquitto_mtls.conf
dashboard_subscriber.py            mosquitto_tls.conf
experiment_runner.py               publisher_mtls.py
generate_certs.py                 sensor_publisher.py
generate_client_certs.py          subscriber_mtls.py
mosquitto_insecure.conf
christinekh@christinekhsMBP Hydroficient-project % ls certs
ca-key.pem          device-002-key.pem  server-key.pem
ca.pem              device-002.pem     server.pem
device-001-key.pem  device-003-key.pem wrong-ca.pem
device-001.pem      device-003.pem
christinekh@christinekhsMBP Hydroficient-project %
```

- Take a screenshot of the script output.

Part2: Configure Mosquitto for mTLS

- Key settings to verify are present:
 - `require_certificate true`
 - `use_identity_as_username true`



The screenshot shows a code editor with a file explorer on the left and a code editor on the right. The file explorer shows a project named 'HYDROFICIENT-PROJECT' with several files and folders. The code editor shows the contents of 'mosquitto_mtls.conf'.

```
mosquitto_mtls.conf
37 # Used to verify BOTH server (by clients) and clients (by server)
38 cafile certs/ca.pem
39
40 # Path to server certificate
41 # This proves the broker's identity to clients
42 certfile certs/server.pem
43
44 # Path to server private key
45 # Keep this file secure - it's the broker's secret
46 keyfile certs/server-key.pem
47
48 # -----
49 # mTLS Settings (The Key Change from Project 4)
50 # -----
51
52 # REQUIRE clients to present a valid certificate
53 # This is THE setting that enables mTLS
54 # Without this: anyone with CA cert can connect (one-way TLS)
55 # With this: only devices with valid client certs can connect (mutual TLS)
56 require_certificate true
57
58 # Use the certificate's Common Name as the client username
59 # This means "HYDROLOGIC-MainBuilding-001" becomes the username
60 # Useful for access control and logging
61 use_identity_as_username true
62
63 # -----
64 # Security Settings
65 # -----
66
67 # Disable anonymous connections
68 # Combined with require_certificate, this ensures all clients are verified
69 allow_anonymous false
70
```

- Start the broker with mTLS configuration:
 - `mosquitto -c mosquitto_mtls.conf -v`
- Verify it started correctly. Look for these messages:

mosquitto version 2.0.x starting

Opening TLS listener on port 8883.

```
Hydroficient-project — mosquitto -c mosquitto_mtls.conf -v — 87x33
Last login: Tue Sep 22 10:21:59 on ttys002
christinekh@christinekh-MBP ~ % cd "/Users/christinekh/Documents/IoT Hydroficient-Extern/Week5/Hydroficient-project"
christinekh@christinekh-MBP Hydroficient-project % pwd
/Users/christinekh/Documents/IoT Hydroficient-Extern/Week5/Hydroficient-project
christinekh@christinekh-MBP Hydroficient-project % ls
certs                                mosquitto_mtls.conf
dashboard_subscriber.py             mosquitto_tls.conf
experiment_runner.py                publisher_mtls.py
generate_certs.py                   sensor_publisher.py
generate_client_certs.py            subscriber_mtls.py
mosquitto_insecure.conf
christinekh@christinekh-MBP Hydroficient-project % ls certs
ca-key.pem          device-002-key.pem    server-key.pem
ca.pem              device-002.pem       server.pem
device-001-key.pem  device-003-key.pem   wrong-ca.pem
device-001.pem      device-003.pem
christinekh@christinekh-MBP Hydroficient-project % mosquitto -c mosquitto_mtls.conf -v
1790063722: Info: running mosquitto as user: christinekh.
1790063722: mosquitto version 2.1.2 starting
1790063722: Config loaded from mosquitto_mtls.conf.
1790063722: Bridge support available.
1790063722: Persistence support available.
1790063722: TLS support available.
1790063722: TLS-PSK support available.
1790063722: Websockets support available.
1790063722: Opening ipv6 listen socket on port 8883.
1790063722: Opening ipv4 listen socket on port 8883.
1790063722: mosquitto version 2.1.2 running
```

Part 3: Run the mTLS Pipeline

- Verify both show successful connection:
 - Subscriber shows `[SUCCESS] Connected to broker`
 - Publisher shows `[SUCCESS] Connected to broker`
 - Subscriber displays received sensor data
- Take screenshots showing:
 - Publisher output with successful connection
 - Subscriber output with received data

```

Terminal Shell Edit View Window Help
Hydroficient-project -- -zsh -- 99x55
Last login: Tue Sep 22 10:19:24 on ttys000
christinekh@christinekh-MBP ~ % cd /Users/christinekh/Documents/IoT\ Hydroficient-Extern/Week5/Hydroficient-project
christinekh@christinekh-MBP Hydroficient-project % python3 subscriber_mtls.py
=====
Grand Marina Security Dashboard (mTLS)
=====
Subscribing to: hydroficient/grandmarina/#
Certificate: certs/device-001.pem
=====
/Users/christinekh/Documents/IoT Hydroficient-Extern/Week5/Hydroficient-project/subscriber_mtls.py:
100: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
client = mqtt.Client(client_id=CLIENT_NAME, **MQTT_CLIENT_ARGS)
[INFO] mTLS configured with client certificate

[CONNECTING] localhost:8883...

[LISTENING] Waiting for messages (Ctrl+C to stop)...

[SUCCESS] Connected to broker as GrandMarina-Dashboard
[INFO] Subscribing to: hydroficient/grandmarina/#
[SUBSCRIBED] QoS granted: (1,)

=====
[RECEIVED] Topic: hydroficient/grandmarina/device-001/sensors
[TIME] 10:22:29
=====
Device: HYDROLOGIC-Device-001
Flow Rate: 53.68 LPM
Pressure (Up): 60.11 PSI
Pressure (Down): 56.95 PSI
Status: operational

=====
[RECEIVED] Topic: hydroficient/grandmarina/device-001/sensors
[TIME] 10:22:34
=====
Device: HYDROLOGIC-Device-001
Flow Rate: 48.02 LPM
Pressure (Up): 59.94 PSI
Pressure (Down): 54.85 PSI
Status: operational

=====
[RECEIVED] Topic: hydroficient/grandmarina/device-001/sensors
[TIME] 10:22:39
=====
Device: HYDROLOGIC-Device-001
Flow Rate: 47.07 LPM
Pressure (Up): 61.3 PSI
Pressure (Down): 56.43 PSI
Status: operational

=====
[RECEIVED] Topic: hydroficient/grandmarina/device-001/sensors
[TIME] 10:22:44
=====
Device: HYDROLOGIC-Device-001
Flow Rate: 47.17 LPM
Pressure (Up): 60.12 PSI
Pressure (Down): 54.19 PSI
Status: operational

=====
[RECEIVED] Topic: hydroficient/grandmarina/device-001/sensors
[TIME] 10:22:49
=====
Device: HYDROLOGIC-Device-001
Flow Rate: 47.05 LPM
Pressure (Up): 60.05 PSI
Pressure (Down): 57.28 PSI
Status: operational

=====

Hydroficient-project -- -zsh -- 104x55
Last login: Tue Sep 22 10:20:13 on ttys001
christinekh@christinekh-MBP ~ % cd /Users/christinekh/Documents/IoT\ Hydroficient-Extern/Week5/Hydroficient-project
christinekh@christinekh-MBP Hydroficient-project % python3 publisher_mtls.py
=====
HYDROLOGIC Sensor Publisher (mTLS)
=====
Device ID: 001
Topic: hydroficient/grandmarina/device-001/sensors
Certificate: certs/device-001.pem
=====
/Users/christinekh/Documents/IoT Hydroficient-Extern/Week5/Hydroficient-project/publisher_mtls.py:108: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
client = mqtt.Client(client_id=CLIENT_NAME, **MQTT_CLIENT_ARGS)
[INFO] mTLS configured with client certificate

[CONNECTING] localhost:8883...
[SUCCESS] Connected to broker as HYDROLOGIC-Device-001
[INFO] Certificate identity verified by broker

[PUBLISHING] Sending sensor readings (Ctrl+C to stop)...

[1] Published: 53.68 LPM
[PUBLISHED] Message ID: 1
[2] Published: 48.02 LPM
[PUBLISHED] Message ID: 2
[3] Published: 47.07 LPM
[PUBLISHED] Message ID: 3
[4] Published: 47.17 LPM
[PUBLISHED] Message ID: 4
[5] Published: 47.05 LPM
[PUBLISHED] Message ID: 5
[6] Published: 46.97 LPM
[PUBLISHED] Message ID: 6
[7] Published: 46.09 LPM
[PUBLISHED] Message ID: 7
[8] Published: 50.34 LPM
[PUBLISHED] Message ID: 8
[9] Published: 48.33 LPM
[PUBLISHED] Message ID: 9
[10] Published: 54.22 LPM
[PUBLISHED] Message ID: 10
[11] Published: 52.61 LPM
[PUBLISHED] Message ID: 11
[12] Published: 46.2 LPM
[PUBLISHED] Message ID: 12
[13] Published: 45.3 LPM
[PUBLISHED] Message ID: 13
[14] Published: 53.41 LPM
[PUBLISHED] Message ID: 14
[15] Published: 47.71 LPM
[PUBLISHED] Message ID: 15
[16] Published: 48.31 LPM
[PUBLISHED] Message ID: 16
[17] Published: 49.87 LPM
[PUBLISHED] Message ID: 17
[18] Published: 45.15 LPM
[PUBLISHED] Message ID: 18
[19] Published: 51.64 LPM
[PUBLISHED] Message ID: 19
[20] Published: 53.15 LPM
[PUBLISHED] Message ID: 20
[21] Published: 51.32 LPM
[PUBLISHED] Message ID: 21
[22] Published: 54.52 LPM
[PUBLISHED] Message ID: 22
[23] Published: 51.04 LPM
[PUBLISHED] Message ID: 23
[24] Published: 54.25 LPM
[PUBLISHED] Message ID: 24

```

Part 4: Verify Rejection Without Certificate

- This is the key test. Verify that your broker rejects connections without valid client certificates.

- Create a copy of the publisher without mTLS credentials:

```
# test_no_cert.py - Should be REJECTED
```

```
import paho.mqtt.client as mqtt
```

```
import ssl
```

```
# Handle paho-mqtt 2.0+ API change
```

```
try:
```

```
    MQTT_CLIENT_ARGS = {"callback_api_version": mqtt.CallbackAPIVersion.VERSION1}
```

```
except AttributeError:
```

```
    MQTT_CLIENT_ARGS = {}
```

```
client = mqtt.Client(client_id="rogue-device", **MQTT_CLIENT_ARGS)
```

```
# Only CA cert, NO client certificate
```

```
client.tls_set(ca_certs="certs/ca.pem")
```

```
try:
```

```
    client.connect("localhost", 8883, keepalive=60)
```

```
    print("ERROR: Connection should have been rejected!")
```

```
except Exception as e:
```

```
print(f"SUCCESS: Connection rejected: {e}")
```

- Run the test:

- `python test_no_cert.py`

- Record the result:

- Connection was rejected (CORRECT BEHAVIOR)
- Connection succeeded (PROBLEM - check your config!)

- Record the error message:

- Take a screenshot of the rejection.

The image shows two terminal windows side-by-side. The left window, titled 'Hydroficient-project -- zsh -- 105x44', shows the execution of 'python3 test_no_cert.py'. It displays a deprecation warning about Callback API version 1, followed by an 'ERROR: Connection should have been rejected!' message. The right window, titled 'Hydroficient-project -- mosquitto -c mosquitto_mtls.conf -v -- 98x55', shows the execution of 'mosquitto -c mosquitto_mtls.conf -v'. It displays various status messages from Mosquitto, including 'Info: running mosquitto as user: christinekh.', 'Config loaded from mosquitto_mtls.conf.', and 'New connection from ::1:51497 on port 8883.'. At the bottom, it shows a 'Disconnected: Protocol error.' message.